Defining the OH quantity in Lawrence et al. (2001)

Reference. Lawrence, Jöckel, and von Kuhlmann, *Atmospheric Chemistry and Physics* 1, 37–49, doi:10.5194/acp-1-37-2001.

Question and method. A global mean OH concentration depends on both the averaging weights and the atmospheric domain. The authors compare four OH distributions: one empirical field and three MATCH chemistry-model fields. They calculate air-mass-weighted, volume-weighted, and reaction-weighted OH means, along with CH_4 and methyl-chloroform (MCF) lifetimes. For a species X, the relevant diagnostic is

$$\tau_X^{-1} = \frac{\sum_i k_X(T_i)[\mathrm{OH}]_i M_{X,i}}{\sum_i M_{X,i}}, \tag{1}$$

in which $M_{X,i}$ is species mass in a model cell and $k_X(T_i)$ is the temperature-dependent reaction rate. Thus a loss frequency is not generally proportional to an arbitrary global OH average. Temperature, species distribution, domain, and averaging convention must match.

Findings. Table 2 gives air-mass-weighted OH means of 1.14, 1.26, 0.88, and 0.95, versus CH_4-reaction-weighted means of 1.32, 1.39, 1.06, and 1.19, all in 10^6 molecules cm^{-3}, for the four fields. Table 3 gives CH_4 lifetimes of 8.23, 7.79, 10.25, and 9.12 years. Weighting choices can change a reported global mean by about 30%; differences in domain add another source of disagreement. More than 60% of methane and MCF oxidation occurs in the tropical lower troposphere, while the contribution above 250 hPa is small. Consequently, a large OH anomaly where oxidation is weak need not strongly change methane lifetime.

Recommended diagnostics. The paper recommends reporting both mass-weighted and reaction-weighted means and a 12-region decomposition with latitude boundaries at 30°S, the equator, and 30°N and pressure boundaries at 750, 500, and 250 hPa. Regional diagnostics expose differences that a single global mean conceals. The paper also distinguishes a species' chemical lifetime from the response time of a perturbation when chemical feedbacks operate.

Use in this reconstruction. This is a methodological reference, not a reconstruction of 2019–2022 methane fluxes. Its tabulated OH means and lifetimes are retained as a numerical illustration of the weighting issue; no 2001 field is inserted into the surge budget. We use annual OH lifetimes or matched mass and removal fluxes when available. Reported area- or mass-weighted OH changes are retained as such, rather than converted silently into equivalent emissions. This decision matters especially when comparing He and Qu's hemispheric OH optimization with Feng's latitude-band optimization and Peng's three-dimensional chemistry. No separate supplementary file is linked by the journal article; the derivations, tables, and regional diagnostics are in the main paper.